RoboPIN: Grounded Embodied Reasoning via Pinned Chain-of-Thought

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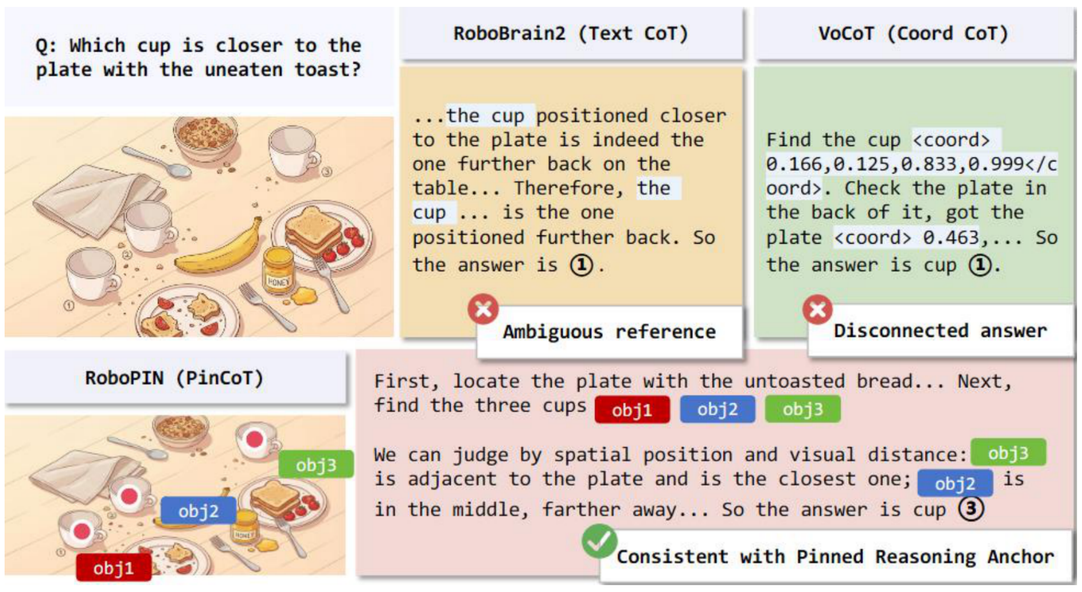
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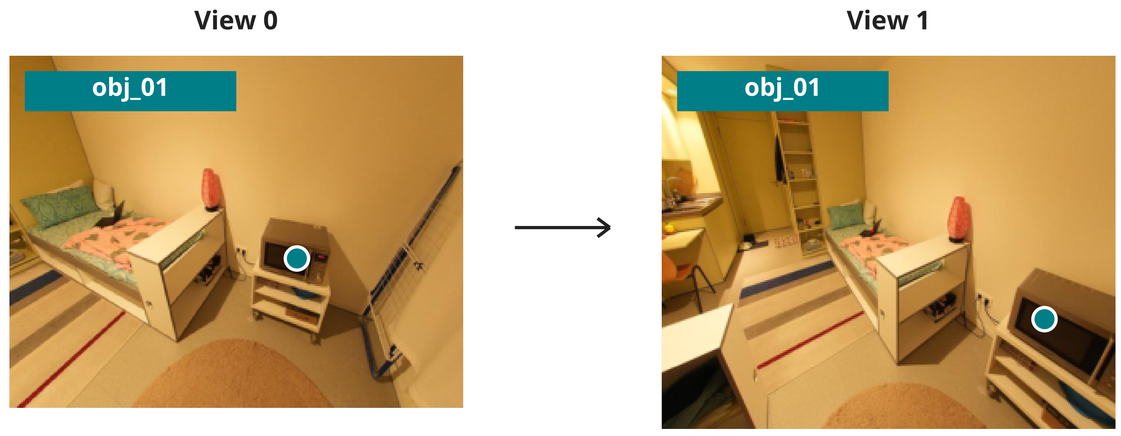
Motivation



1. Text-based references can drift between similar objects. Coordinate-only grounding lacks explicit identity tracking.

2. A model may correctly analyze object A, then silently point to object B. This reasoning-to-answer misalignment breaks reliable visual grounding.

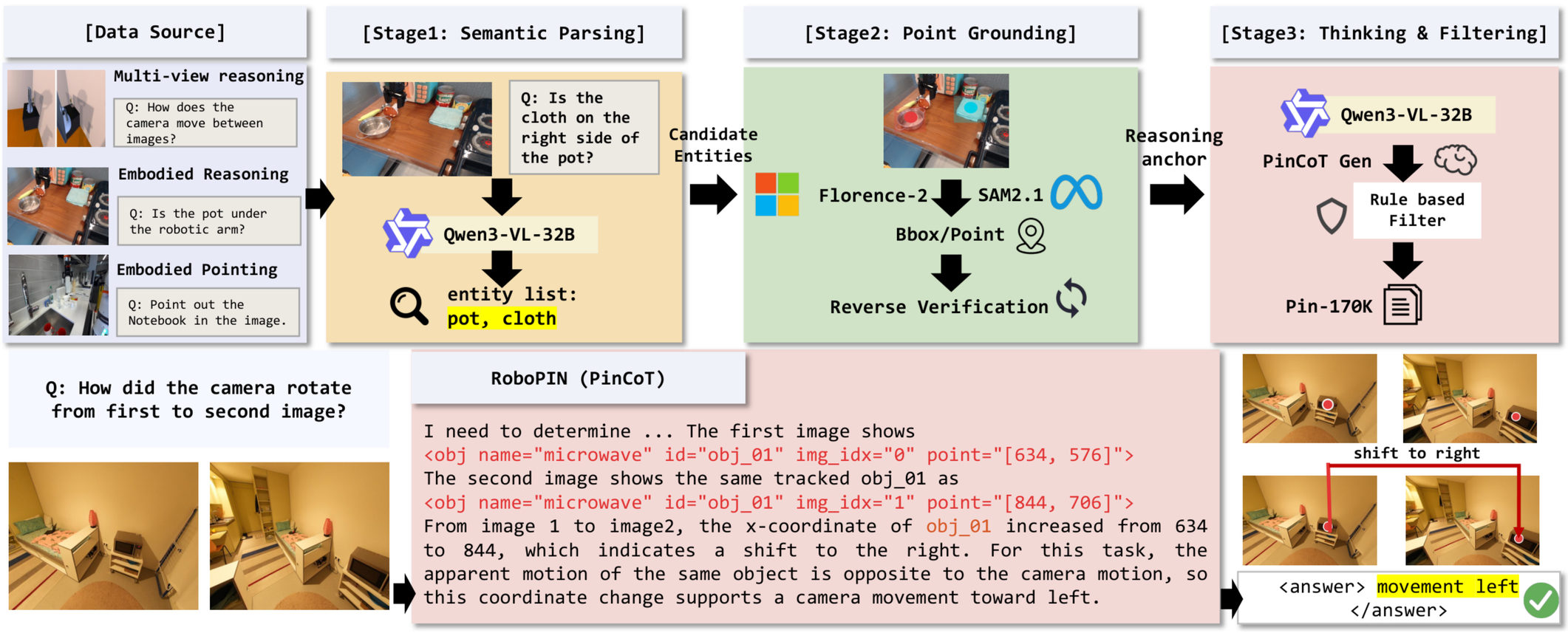
Key Insight: Persistent Identity



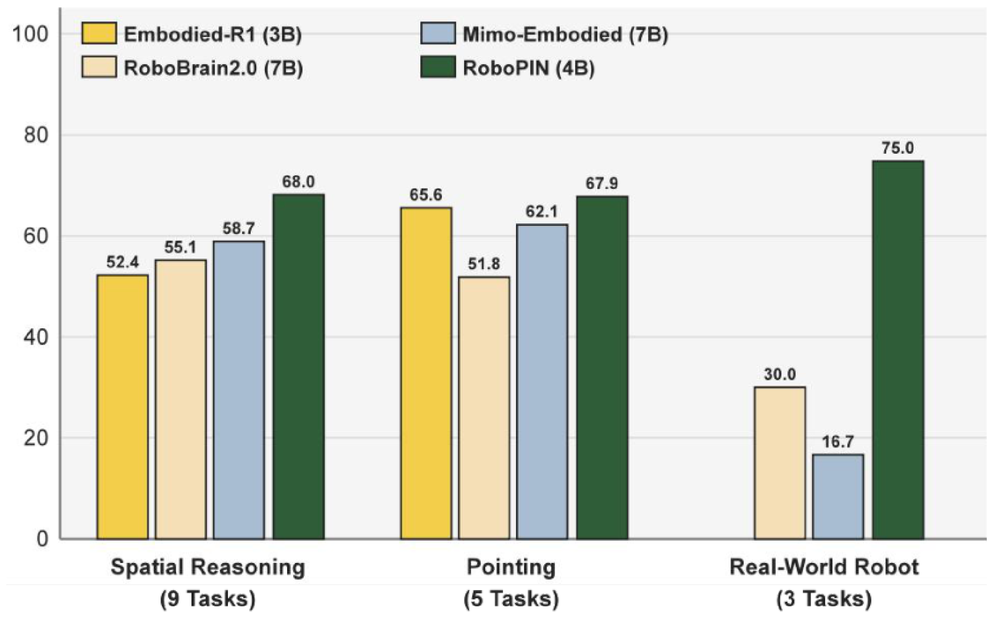
From isolated grounding to persistent anchors

- **Spatial references are usually isolated annotations, without explicit identity tracking during reasoning.**
- **PinCoT binds each entity to a name, ID, view and point. Later steps reuse the same ID.**
- **Our insight: visual anchors act as persistent reasoning variables, making each step traceable to visual evidence.**

Our Method: RoboPIN / PinCoT



Experimental Results



Ablation studies

Reward design

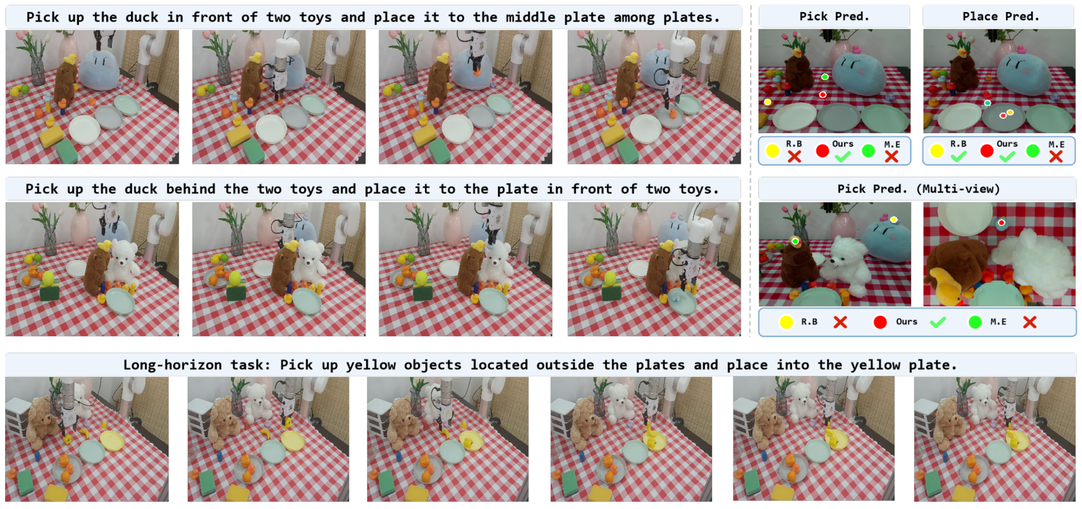
Setting	EmbSpatial
Full reward	**84.7**
w/o Pin	82.0
w/o Consistency	82.7

Training stages

Stage	Spatial	Point
SFT	57.0	66.0
+ CoT-SFT	62.0	63.6
+ RFT	**67.7**	**67.9**

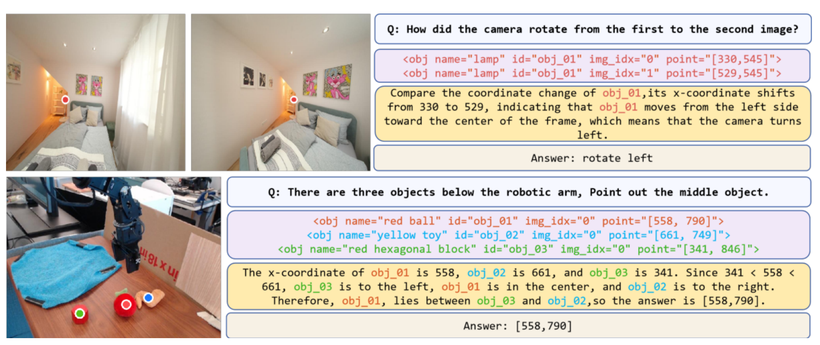
Spatial reasoning: +10.7 points from SFT to RFT.

Qualitative Results



Zero-shot xArm · successes / 20

20 / 20	14 / 20	11 / 20
Spatial	Multi-view	Long-horizon



Persistent Visual Anchors · Consistent Reasoning · Grounded Answers